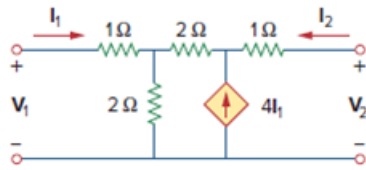


Problem

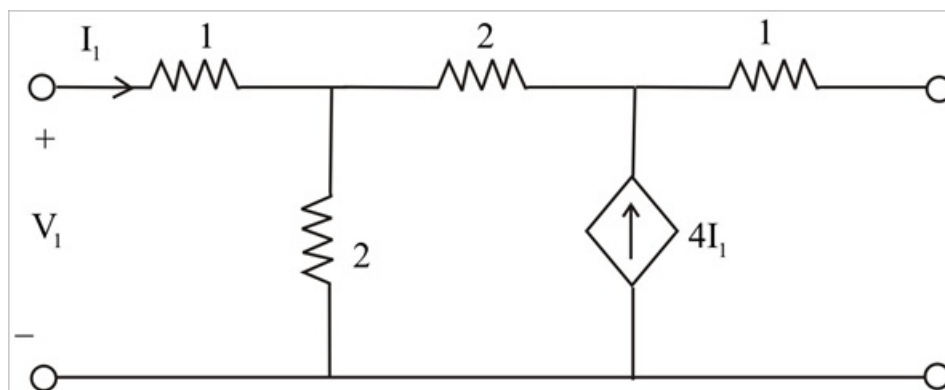
Determine the hybrid parameters for the network in Fig. 19.89.

Figure 19.89



Step-by-step solution

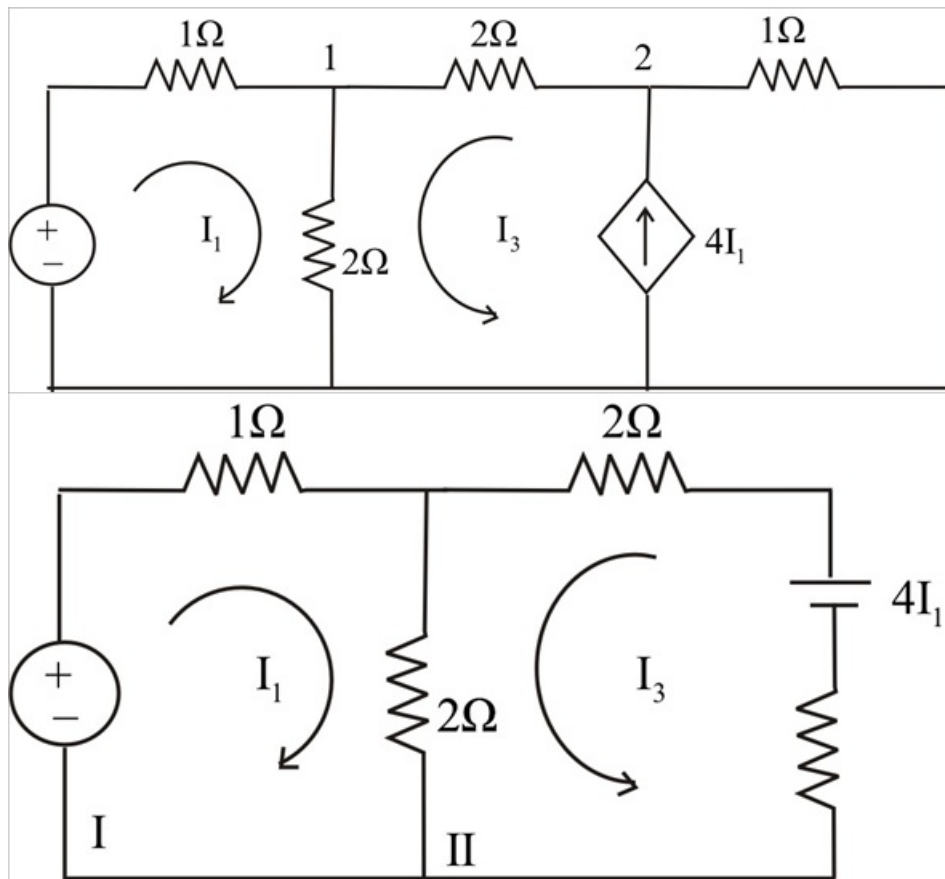
Step 1 of 6



Step 2 of 6

$$V_2 = 0$$

we have



Step 3 of 6

$$V_1 = 3I_1 + 2I_2 \rightarrow \text{Loop I}$$

$$4I_1 = 5I_2 + 2I_1 \rightarrow \text{Loop II}$$

$$5 I_2 - 2I_1 = 0 \quad -(2)$$

$$I_1 = \frac{5I_2}{2}$$

$$2 \quad V_1 = 6I_1 + 4I_2$$

$$0 \quad = -6I_1 + 15I_2$$

$$= \frac{10V_1}{2 \times 19} = \frac{5}{19} V_1$$

$$\begin{array}{r} + \quad + \\ \hline 2V_1 \quad = 19I_2 \end{array}$$

Step 4 of 6

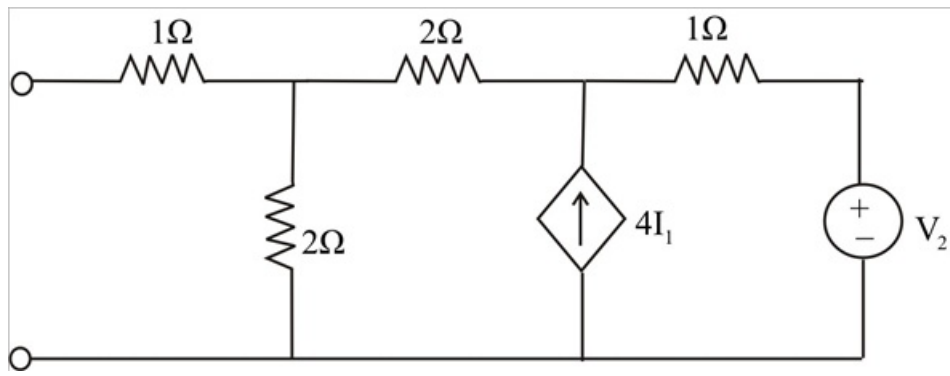
$$\frac{V_1}{I_1} = \frac{19}{5} = 3.8\Omega$$

$$I_2 = \frac{2}{19} V_1 = \frac{2}{19} \times 3.8 I_1$$

$$\frac{I_2}{I_1} = 0.4.$$

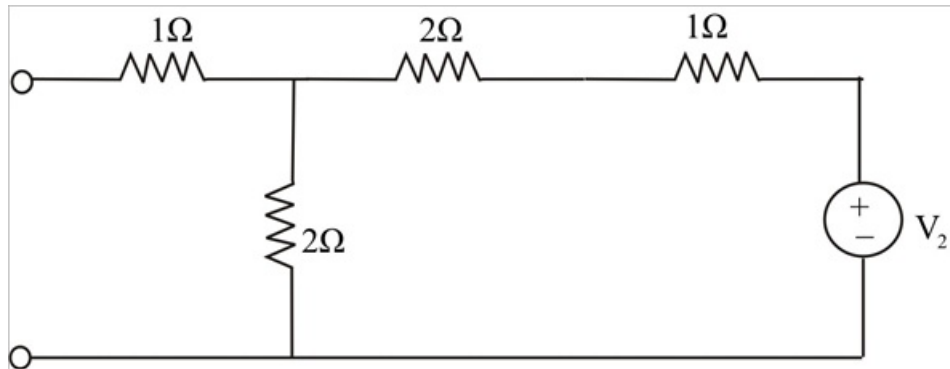
$$I_1 = 0 \quad \text{we have}$$

Step 5 of 6



Step 6 of 6

since $I_1 = 0$ we have $4I_1$ current source as Open circuited



$$\frac{V_2}{I_2} = \frac{V_2}{\frac{V_2}{5}} = 5\Omega \quad \frac{I_2}{V_2} = 0.21S$$

$$\frac{V_1}{V_2} = \frac{2 \times \frac{V_2}{5}}{V_2} = 0.4 \quad \begin{bmatrix} 3.8\Omega & 0.4 \\ -0.4 & 0.2S \end{bmatrix}$$